Mohamed Abdalla Yahya Ahmed

MECHATRONICS ENGINEERING STUDENT

Malaysia, Kuala Lumpur | https://mhmd-a-y.github.io/my-portfolio/

Third-year Mechatronics Engineering student at UCSI University, currently in Semester 2, with hands-on industrial exposure to crane maintenance, PLCs, drive systems, and control systems. Experienced in engineering projects involving embedded systems, microcontrollers, automation simulation, and sensor integration. Seeking an internship to apply practical engineering knowledge, strengthen hardware and software skills, and contribute to real-world engineering projects in automation, robotics, or R&D.

EDUCATION

UCSI University — Kuala Lumpur, Malaysia
Bachelor of Mechatronics Engineering with Honours | May 2024 – Present
CGPA: 3.52 | Dean's List, January 2025

INDUSTRIAL EXPERIENCE

Sea Ports Corporation (SPC) — Port Sudan, Sudan
Engineering Intern, Cranes Department | January 2026 – May 2026

- Supported maintenance and troubleshooting activities involving Ship-to-Shore (STS) and Rubber-Tyred Gantry (RTG) cranes in an industrial port environment.
- Gained practical exposure to PLC-based control systems, variable-frequency drives (VFDs), encoders, PROFIBUS networks, and electrical control components.
- Used Siemens STEP 7, DriveMonitor, Fuji PLC software, and crane monitoring systems to examine equipment status and support fault investigation.
- Participated in maintenance and electrical testing activities, including insulation resistance testing, while following industrial safety practices.

SELECTED ENGINEERING PROJECTS

OMRON PLC Programming and HMI Development | Fluid Power & Drives

- Developed a Ladder Logic control program for an OMRON PLC using simulation.
- Created an HMI interface to support operator interaction with the simulated control system.

Smart Temperature and Distance Monitoring System | PIC18F452

- Programmed a PIC18F452 microcontroller to integrate temperature and ultrasonic distance measurements.
- Implemented sensor input processing, LCD output, and threshold-based control logic for a monitoring system.

Single-Axis Solar Tracking System | Academic Project

- Developed a solar-tracking prototype using Arduino, light sensors, a servo motor, and a solar panel.
- Integrated sensor feedback, control logic, and mechanical components to adjust panel orientation according to light levels.

Hydraulic and Pneumatic Sequence Control | Fluid Power & Drives

- Designed and simulated hydraulic and pneumatic sequence-control circuits in FluidSIM to study actuator sequencing and system operation.

Edge AI and Computer Vision for Automated Control | Ongoing Research

- Researching the design of an integrated system combining computer vision, edge AI, and automated control.
- Exploring system architecture and the potential integration of intelligent perception with control functions.

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

UCSI Institute of Engineering & Technology (IET) | Committee Member
March 2025 – Present

- Supported engineering events and technical workshops through event coordination and publicity.
- Helped organise and deliver two campus web development workshops for fellow students.

Sudanese Students Society (SSS), UCSI | Secretary & Student Affairs Committee
May 2025 – Present

- Managed committee records and supported communication and coordination across a committee of more than 30 members.
- Contributed to student support initiatives, including a mental health workshop, a student guide for new students, and an English Club.

Additional Qualifications: Harvard CS50x: Introduction to Computer Science (2023); FPGA Implementation with Intel Quartus and Verilog HDL Workshop (2025); PCB Design Workshop (2025).

Additional Experience: Freelance full-stack web development, including client websites and web applications using PHP, Laravel, JavaScript, MySQL, and WordPress.

SKILLS

Technical Skills

- **Industrial Automation & Controls:** OMRON PLC programming (simulation), Ladder Logic, HMI interface development, Siemens STEP 7, Fuji PLC software, DriveMonitor, VFDs, PROFIBUS, FluidSIM.
- **Embedded Systems & Electronics:** PIC18F452, Arduino, C programming, sensor integration, microcontroller-based monitoring systems, electrical measurement.
- **CAD & Engineering Design:** SolidWorks, AutoCAD, hydraulic and pneumatic circuit design and simulation.
- **Programming & Software:** C, Python, PHP, Laravel, JavaScript, MySQL, MATLAB (basic).
- **Engineering Competencies:** Troubleshooting, control-system fundamentals, electromechanical integration, technical documentation, teamwork.

Other Skills:

Problem-solving **|** Critical & analytical thinking **|** Time management **|** Event Management **|** Public Speaking | Leadership | Adaptability | Troubleshooting mindset.

LANGUAGES

Arabic (Native) | English (Fluent)

REFEREES

- Asst. Prof. Ts. Dr. Cik Suhana Binti Hassan (Head of Department, UCSI University). suhana@ucsiuniversity.edu.my
- Dr. Wan Zailah Wan Said (Lecturer, UCSI University). Zailah@ucsiuniversity.edu.my